

Lecture 26/Module 3

Sinusoidal Steady State Analysis

12/09/2026

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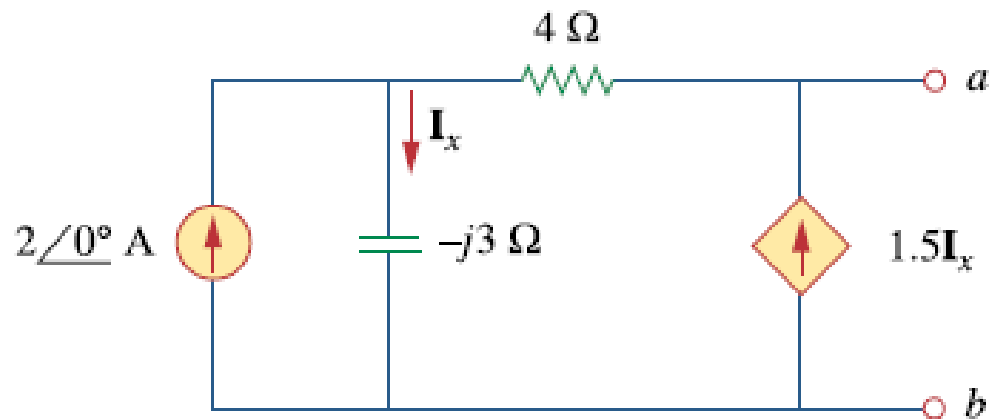
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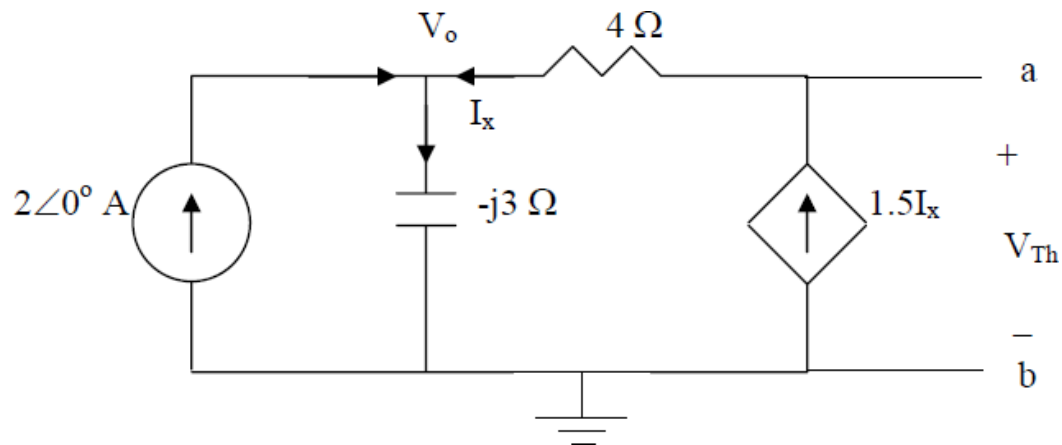
Exercise 1

Find the Thevenin equivalent at terminals a - b of the circuit



Answer

To find V_{Th} , consider the circuit below



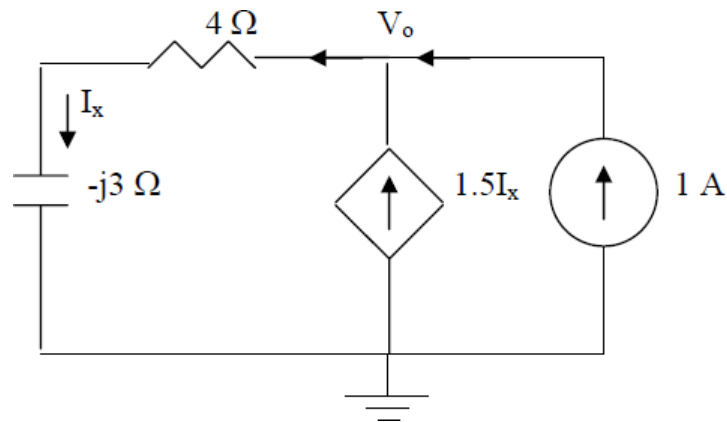
$$2 + 1.5I_x = I_x \longrightarrow I_x = -4$$

$$\text{But } V_o = -j3I_x = j12$$

$$V_{Th} = V_o + 6I_x = \underline{j12 - 24\text{ V}}$$

Answer

To find Z_{Th} , consider the circuit shown below.



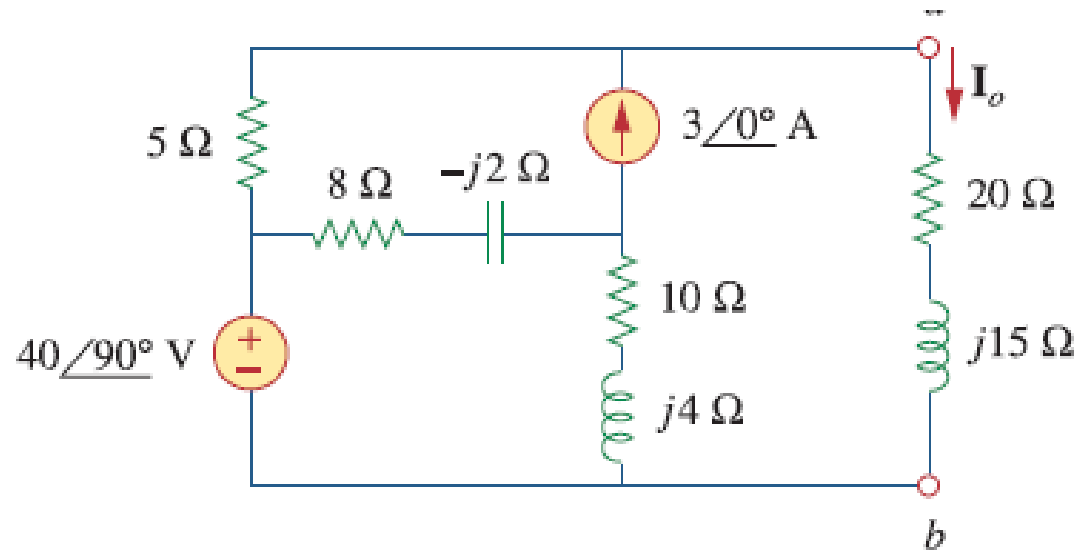
$$1 + 1.5 I_x = I_x \Rightarrow I_x = -2$$

$$-V_o + I_x(4 - j3) = 0 \longrightarrow V_o = -8 + j6$$

$$Z_{Th} = \frac{V_o}{1} = \underline{\underline{-8 + j6 \, \Omega}}$$

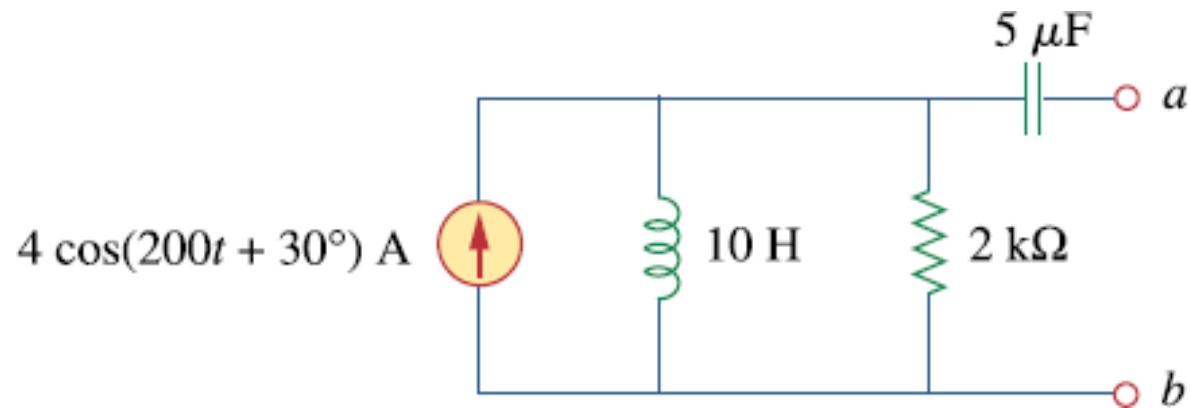
Numerical -1

Obtain current $\mathbf{I}_o$ in Fig. using Norton's theorem.



Exercise -2

- Obtain the Norton's equivalent for the circuit.



Answer

- Solution:

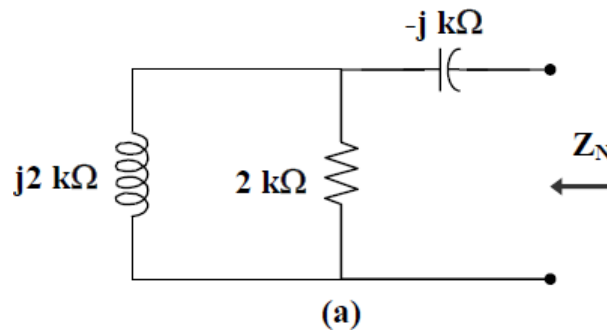
Transform the circuit to the frequency domain.

$$4 \cos(200t + 30^\circ) \longrightarrow 4 \angle 30^\circ, \quad \omega = 200$$

$$10 \text{ H} \longrightarrow j\omega L = j(200)(10) = j2 \text{ k}\Omega$$

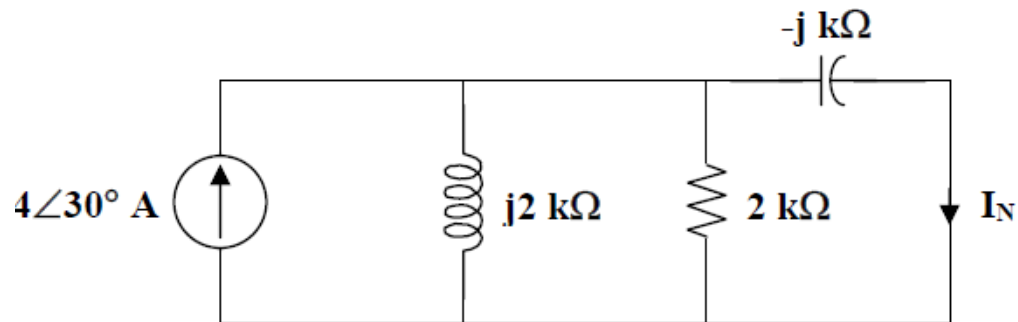
$$5 \mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(200)(5 \times 10^{-6})} = -j \text{ k}\Omega$$

Z_N is found using the circuit in Fig. (a).



$$Z_N = -j + 2 \parallel j2 = -j + 1 + j = 1 \text{ k}\Omega$$

Answer



$$j2 \parallel 2 = 1 + j$$

By the current division principle,

$$\mathbf{I_N} = \frac{1 + j}{1 + j - j} (4\angle 30^\circ) = 5.657\angle 75^\circ$$

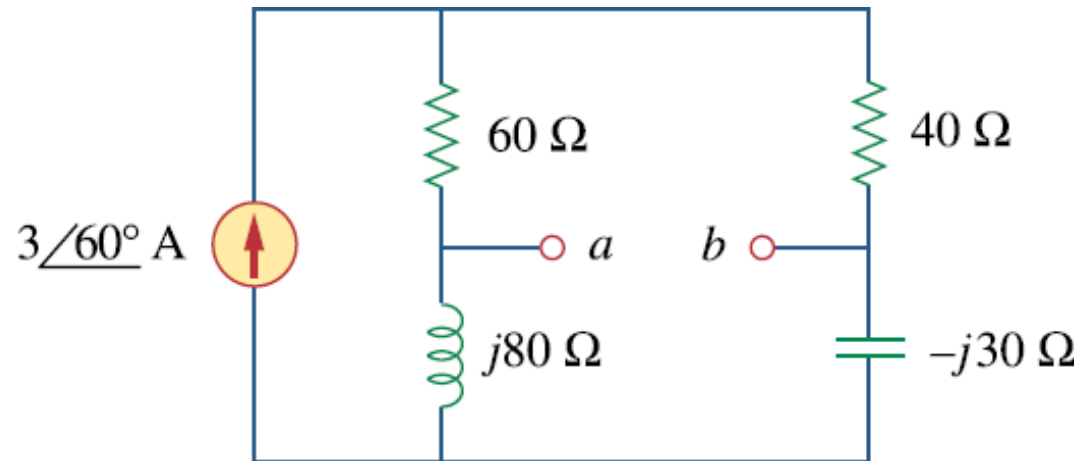
Therefore,

$$\mathbf{i_N} = \underline{\underline{5.657 \cos(200t + 75^\circ) \text{ A}}}$$

$$\mathbf{Z_N} = \underline{\underline{1\text{ k}\Omega}}$$

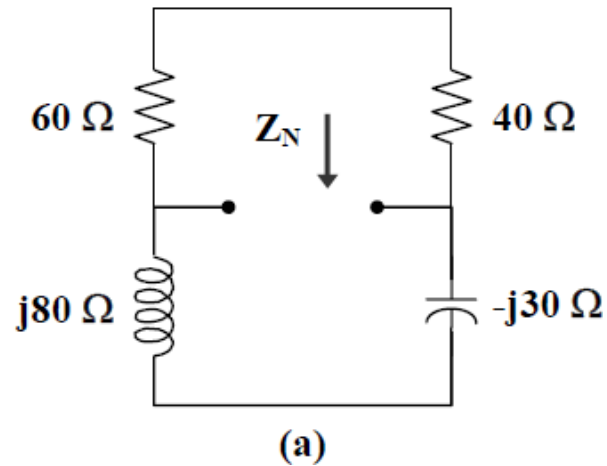
Exercise -3

- Obtain the Norton's equivalent for the circuit.



Answer

- Solution: Z_N is obtained from the circuit in Fig. (a).

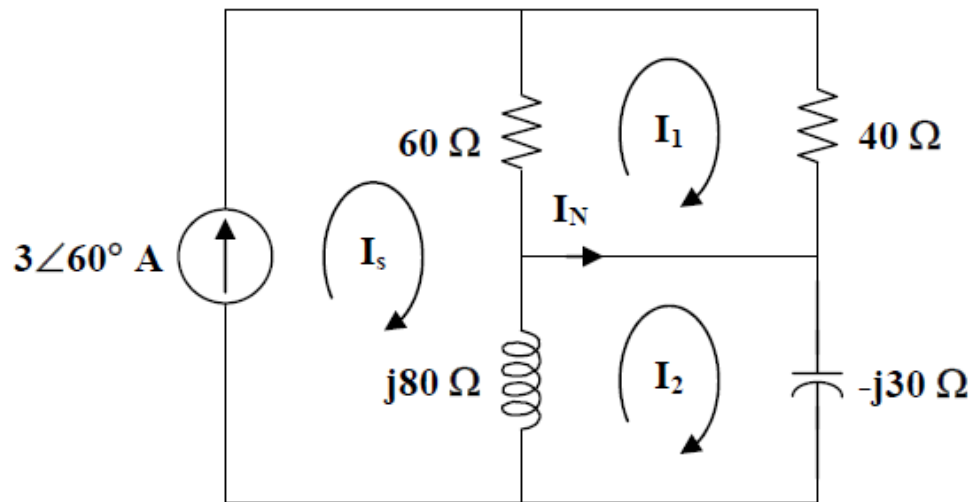


$$Z_N = (60 + 40) \parallel (j80 - j30) = 100 \parallel j50 = \frac{(100)(j50)}{100 + j50}$$

$$Z_N = 20 + j40 = \underline{44.72 \angle 63.43^\circ \Omega}$$

Answer

To find $\mathbf{I}_N$, consider the circuit in Fig. (b).



(b)

$$\mathbf{I}_s = 3\angle 60^\circ$$

For mesh 1,

$$100\mathbf{I}_1 - 60\mathbf{I}_s = 0$$

$$\mathbf{I}_1 = 1.8\angle 60^\circ$$

For mesh 2,

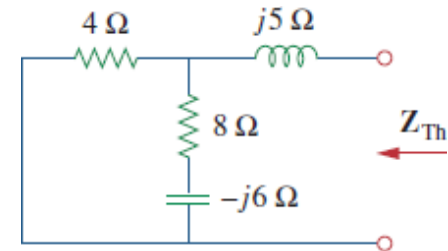
$$(j80 - j30)\mathbf{I}_2 - j80\mathbf{I}_s = 0$$

$$\mathbf{I}_2 = 4.8\angle 60^\circ$$

$$\mathbf{I}_N = \mathbf{I}_2 - \mathbf{I}_1 = \underline{\underline{3\angle 60^\circ \text{ A}}}$$

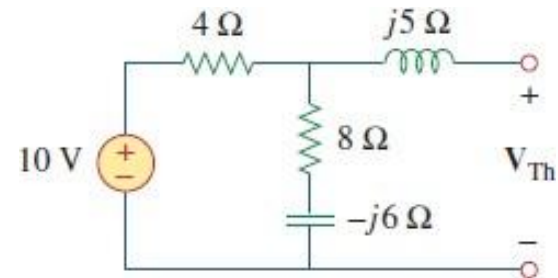
To find Z_{Th} :

$$Z_{Th} = j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} \\ = 2.933 + j4.467 \Omega$$



To find V_{Th} :

$$V_{Th} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \text{ V}$$



The load impedance draws the maximum power from the circuit when

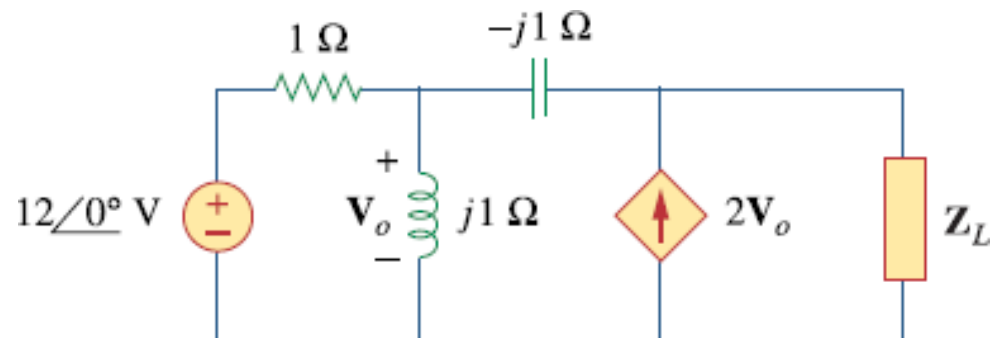
$$Z_L = Z_{Th}^* = 2.933 - j4.467 \Omega$$

the maximum average power is

$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \text{ W}$$

Exercise - 4

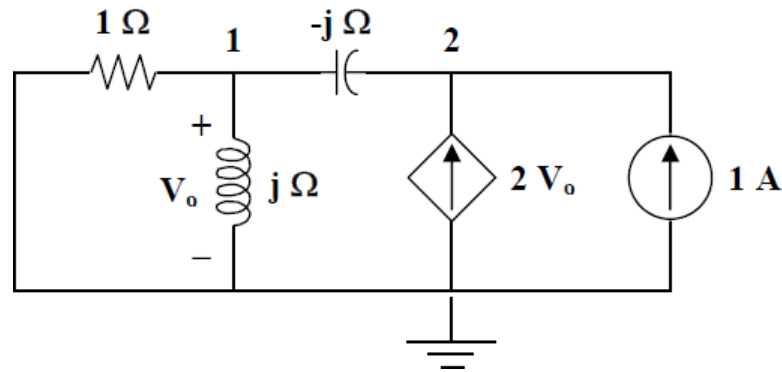
In the circuit of Fig. find the value of $\mathbf{Z}_L$ that will absorb the maximum power and the value of the maximum power.



Answer

- Solution:

To find $\mathbf{Z}_{Th}$, insert a 1-A current source at the load terminals



At node 1,

$$\frac{\mathbf{V}_o}{1} + \frac{\mathbf{V}_o}{j} = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow \mathbf{V}_o = j\mathbf{V}_2 \quad (1)$$

At node 2,

$$1 + 2\mathbf{V}_o = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow 1 = j\mathbf{V}_2 - (2 + j)\mathbf{V}_o \quad (2)$$

Answer

Substituting (1) into (2),

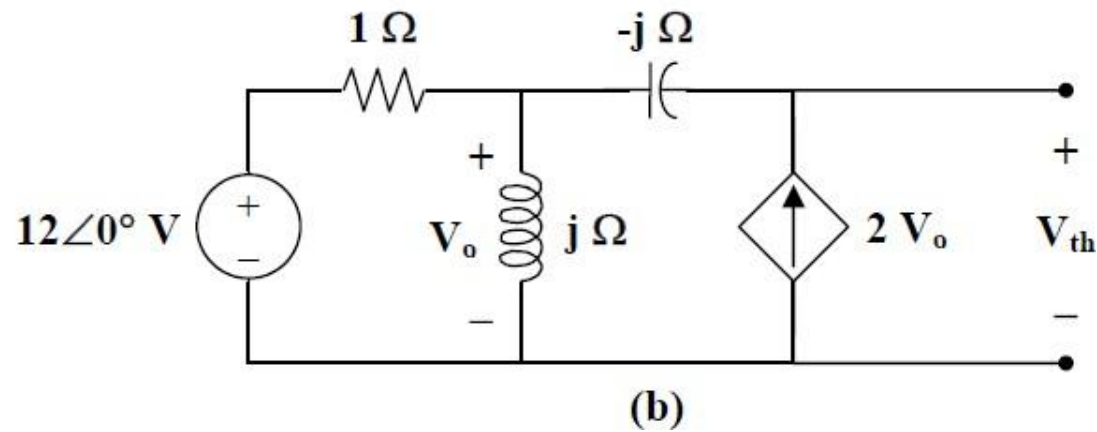
$$1 = j\mathbf{V}_2 - (2 + j)(j)\mathbf{V}_2 = (1 - j)\mathbf{V}_2$$

$$\mathbf{V}_2 = \frac{1}{1 - j}$$

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_2}{1} = \frac{1 + j}{2} = 0.5 + j0.5$$

$$\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = \underline{\underline{0.5 - j0.5 \, \Omega}}$$

We now obtain $\mathbf{V}_{\text{Th}}$ from Fig. (b).



Answer

$$2\mathbf{V}_o + \frac{12 - \mathbf{V}_o}{1} = \frac{\mathbf{V}_o}{j}$$

$$\mathbf{V}_o = \frac{-12}{1+j}$$

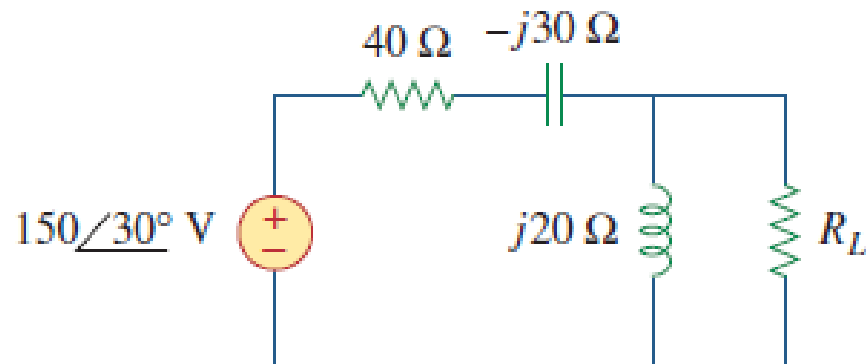
$$-\mathbf{V}_o - (-j \times 2\mathbf{V}_o) + \mathbf{V}_{Th} = 0$$

$$\mathbf{V}_{Th} = (1 - j2)\mathbf{V}_o = \frac{(-12)(1 - j2)}{1 + j}$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_L} = \frac{\left(\frac{12\sqrt{5}}{\sqrt{2}}\right)^2}{(8)(0.5)} = \underline{\underline{90 \text{ W}}}$$

Exercise - 5

Find the value of R_L that will absorb the maximum average power.
Calculate the power.



Answer

Solution:

$$\mathbf{Z}_{\text{Th}} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{j20 + 40 - j30} = 9.412 + j22.35 \, \Omega$$

By voltage division, $\mathbf{V}_{\text{Th}} = \frac{j20}{j20 + 40 - j30} (150 \angle 30^\circ) = 72.76 \angle 134^\circ \text{ V}$

The value of R_L that will absorb the maximum average power is

$$R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{9.412^2 + 22.35^2} = 24.25 \, \Omega$$

The current through the load is

$$\mathbf{I} = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + R_L} = \frac{72.76 \angle 134^\circ}{33.66 + j22.35} = 1.8 \angle 100.42^\circ \text{ A}$$

The maximum average power absorbed by R_L is

$$P_{\text{max}} = \frac{1}{2} |\mathbf{I}|^2 R_L = \frac{1}{2} (1.8)^2 (24.25) = 39.29 \text{ W}$$

Thank You